INTEGRAL TEST AND ESTIMATES OF SUMS

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Name: Solutions

Use the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ to answer the following questions.

1. Show that the function $f(x) = 1/x^2$ satisfies the hypotheses of the Integral Test. For n a fixed integer, compute the improper integral

$$\int_n^{\infty} \frac{1}{x^2}\,\mathrm{d}x.$$

Use this to conclude that the series $\sum_{n=1}^{\infty} 1/n^2$ converges.

$f(x)$ is continuous on $(1,\infty)$

$f'(x) = \frac{d}{dx} x^{-2} = -2x^{-3} = \frac{-2}{x^3} < 0$ on $(1,\infty)$

$\Rightarrow f$ decreasing on $(1,\infty)$.

$\lim_{x\to\infty} \frac{1}{x^2} = 0$

$$\int_n^{\infty} \frac{1}{x^2}dx = \lim_{t\to\infty} \int_n^t x^{-2}dx = \lim_{t\to\infty} \frac{x^{-1}}{-1}\Big|_n^t = \lim_{t\to\infty}\left(\frac{-1}{t} - \left(\frac{-1}{n}\right)\right) = \frac{1}{n}$$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the Integral Test because

$$\int_1^{\infty} \frac{1}{x^2}dx = \frac{1}{1} = 1 < \infty.$$

2. Use a calculator to find

$$s_{10} = \sum_{n=1}^{10} \frac{1}{n^2}.$$

Use the intequality

$$\int_{11}^{\infty} \frac{1}{x^2}\,\mathrm{d}x \le R_{10} \le \int_{10}^{\infty} \frac{1}{x^2}\,\mathrm{d}x$$

to determine how good this estimate to the sum of the series is.

$$S_{10} = \sum_{n=1}^{10} \frac{1}{n^2} \approx 1.55$$

$$\frac{1}{11} = \int_{11}^{\infty} \frac{1}{x^2}dx \le R_{10} \le \int_{10}^{\infty} \frac{1}{x^2}dx = \frac{1}{10}$$

$\Rightarrow$ S_{10} is correct to at least one decimal place.

3. Use the inequality

$$s_{10} + \int_{11}^{\infty} \frac{1}{x^2}\,\mathrm{d}x \le \sum_{n=1}^{\infty} \frac{1}{n^2} \le s_{10} + \int_{10}^{\infty} \frac{1}{x^2}\,\mathrm{d}x$$

to find an open interval containing the number s. Compute the midpoint of this interval. Is the midpoint a better or worse approximation to the sum of the series than you found in Problem 2? Why or why not?

$$S_{10} + 1/11 \le S \le S_{10} + \frac{1}{10}$$

The midpoint of this interval is

$$\frac{\left(S_{10} + \frac{1}{11}\right) + \left(S_{10} + \frac{1}{10}\right)}{2} = S_{10} + \frac{\frac{10}{110} + \frac{11}{110}}{2}$$

$$= S_{10} + \frac{21}{220}$$

$$\approx 1.6452$$

This is a better approximation because the error is at most half the width of the interval containing s:

$$\frac{\frac{1}{10} - \frac{1}{11}}{2} = \frac{11-10}{2(110)} = \frac{1}{220}$$

4. It is known that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Use this to compare your estimates from Problems 2 and 3.

$$\left|\frac{\pi^2}{6} - S_{10}\right| \approx 0.0951$$

$$\left|\frac{\pi^2}{6} - \left(S_{10} + \frac{21}{220}\right)\right| \approx 0.0002$$

5. Find the number of terms that you would need to ensure an estimate that is accurate to the first 3 decimal places.

$$R_n \leq \frac{1}{n} < \frac{1}{10^3}$$

$$\Rightarrow n > 10^3 = 1000 \text{ terms.}$$

Determine whether the following series converge or diverge.

6. $\sum_{n=1}^{\infty} \frac{2}{5n-1}$

$$\int_1^{\infty} \frac{2}{5x-1}\,dx = \lim_{t\to\infty} 2\int_1^{t} \frac{1}{5x-1}\,dx = \lim_{t\to\infty} \frac{2}{5}\Big[\ln(5x-1)\Big]_1^t$$

$$= \lim_{t\to\infty} \frac{2}{5}\Big(\ln(t-1) - \ln(4)\Big)$$

$$= \infty$$

Diverges by the Integral Test.

7. $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

$$\int_1^{\infty} \frac{x}{x^2+1} = \lim_{t\to\infty} \int_1^{t} \frac{x}{x^2+1} = \lim_{t\to\infty} \int_2^{t^2+1} \frac{1}{u}\,du = \lim_{t\to\infty} \ln(t^2+1) - \ln(2) = \infty$$

$u = x^2+1$

$du = 2x\,dx$

$\Rightarrow \frac{1}{2}du = x\,dx$

$\Rightarrow \sum_{n=1}^{\infty} \frac{n}{n^2+1}$ diverges by the integral test.

8. $\sum_{n=1}^{\infty} n^2 e^{-n^3}$

$u = -x^3 \Rightarrow -\frac{1}{3}du = -\frac{1}{3}(-3x^2)dx$
$= x^2 dx$

$$\int_1^{\infty} x^2 e^{-x^3} dx = \lim_{t\to\infty} \int_1^t x^2 e^{-x^2} dx$$

$$= \lim_{t\to\infty} -\frac{1}{3} \int_{-1}^{-t^3} e^u du$$

$$= \lim_{t\to\infty} -\frac{1}{3}\left(e^{-t^3} - e^{-1}\right)$$

$$= -\frac{1}{3}\left(0 - \frac{1}{e}\right) = \frac{1}{3e} < \infty$$

$\Rightarrow \sum n^2 e^{-n^3}$ converges by the Integral Test.

9. $\sum_{n=1}^{\infty} \frac{1}{n^2+4}$

$u = \frac{x}{2} \Rightarrow 2du = dx$

$$\int_1^{\infty} \frac{dx}{x^2+4} = \lim_{t\to\infty} \int_1^t \frac{1}{4\left(\frac{x^2}{4}+1\right)} dx = \lim_{t\to\infty} \frac{1}{4} \int_{\frac{1}{2}}^{t/2} \frac{1}{u^2+1} 2du$$

$$= \lim_{t\to\infty} \frac{2}{4} \arctan(u) \Big|_{1/2}^{t/2}$$

$$= \lim_{t\to\infty} \frac{1}{2}\left(\arctan(t/2) - \arctan\left(\frac{1}{2}\right)\right)$$

$$= \frac{1}{2}\left(\frac{\pi}{2} - \arctan\left(\frac{1}{2}\right)\right) < \infty$$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converges by the integral test.